



END OF OPENID IN FEDORA

MICHAL KONEČNÝ

Principal Software
Engineer

✉ zlopez@fedoraproject.org

[m] [@zlopez:fedora.im](https://matrix.to/#/@zlopez:fedora.im)

AGENDA

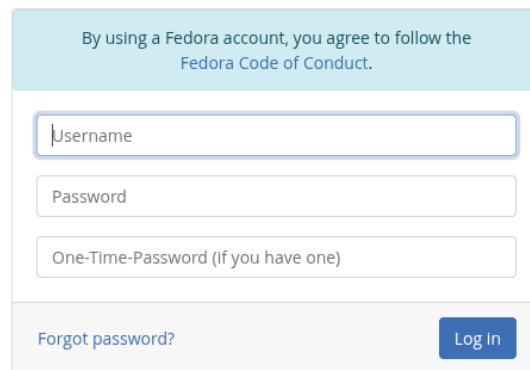
- Background
- Why we did it
- Plan
- Reality
- Current situation
- Future
- Bonus: Introducing new OIDC service

BACKGROUND 1/2

Ipsilon

- Fedora authentication system
- Created 11 years ago
- Connected to identity server (FreeIPA)
- Support for OIDC, SAML2, Kerberos
- <https://pagure.io/ipsilon>

Fedora Community Blog wants to use your Fedora Account
enter your Fedora Account credentials below to authorize **Fedora Community Blog**



The screenshot shows a login form with a light blue header bar containing the text: "By using a Fedora account, you agree to follow the Fedora Code of Conduct." Below the header are three input fields: "Username", "Password", and "One-Time-Password (if you have one)". At the bottom left is a link "Forgot password?" and at the bottom right is a blue "Log In" button.

Ipsilon authentication prompt

BACKGROUND 2/2

OpenID

- Open standard
- Decentralized
- OpenID 2.0 was finalized and published in 2007
- Depreciated and replaced by OIDC



OpenID logo

WHY WE DID IT

Replacing epsilon

- Epsilon is old (11 years)
- Epsilon is unmaintained (we are trying our best to keep it running)
- FreeIPA solutions doesn't support OpenID anymore

OpenID is deprecated

- OpenID is replaced by OIDC
- Most of Fedora is already using OIDC
- OpenID is almost 20 years old



PLAN 1/5

❗ #10241 Port apps to OIDC

Closed: Fixed with Explanation 7 months ago by zlopez. **Opened** 4 years ago by zlopez.

Describe what you would like us to do:

We still have apps that are not yet using OIDC for authentication. It would be nice to add support to them, so we are no longer blocked by this tech debt.

Here is the list of the apps that still needs to be ported:

Apps CPE owns and are critical

pagure (Already implemented, but needs changes in Ipsilon. See <https://pagure.io/fedora-infrastructure/issue/7377> for more details)

mirrormanager

bodhi

noggin

PDC (PDC will be probably retired in future)

FMN

Apps CPE hosts and are critical

Mallman3 / HK

OSBS

* *MBS*

Apps CPE hosts and aren't critical

COPR

nuancier

* *testdays*

It's possible that some of them are already ported over to OIDC (especially those we don't host), it needs to be checked.

When do you need this to be done by? (YYYY/MM/DD)

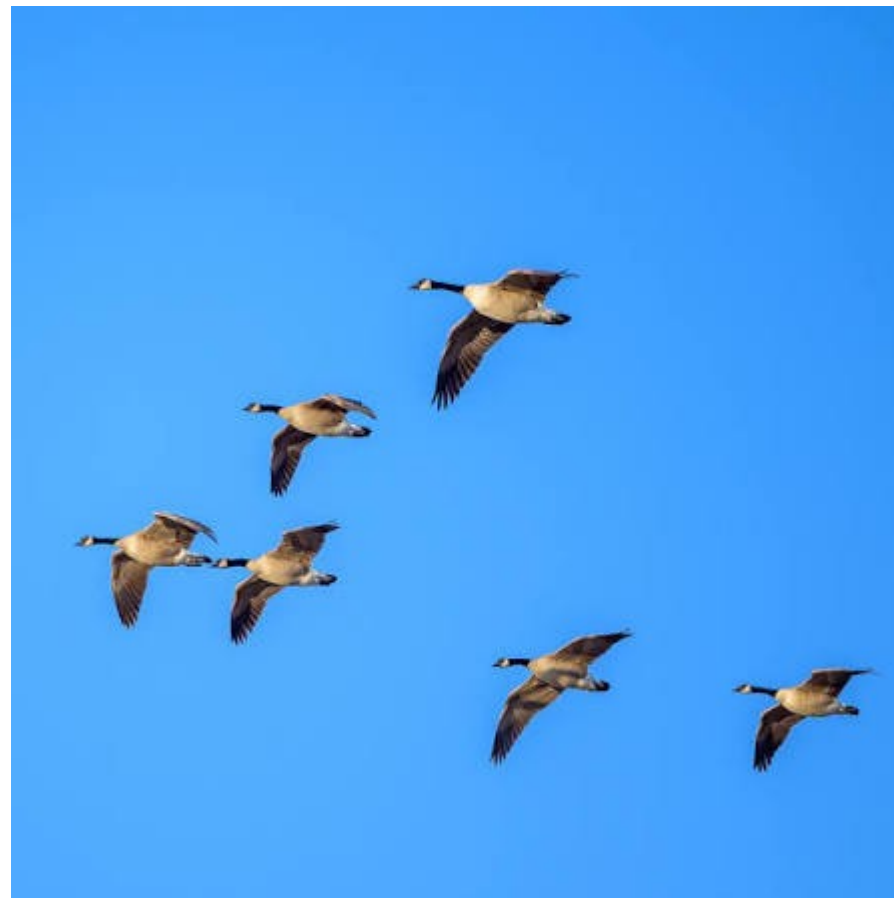
Not urgent.



PLAN 2/5

Migrate apps to OIDC

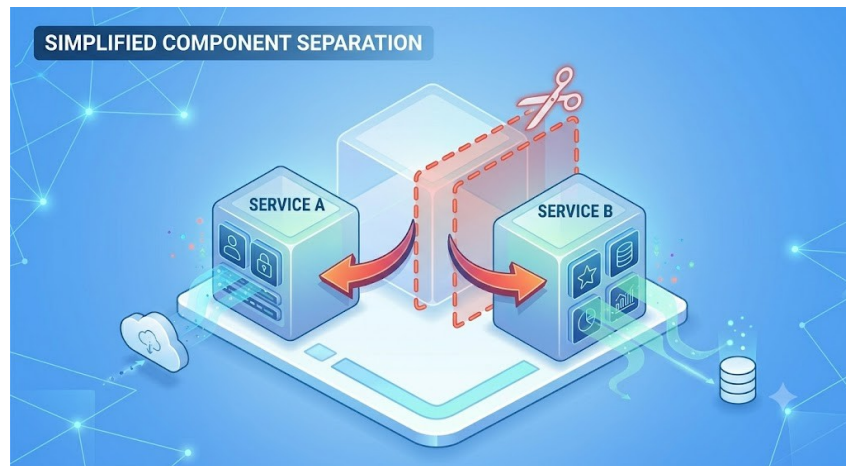
- We had a lot of apps that weren't supporting OIDC
- Critical apps
- Some of them hard to maintain anymore



PLAN 3/5

Separating OpenID

- Create a new OpenID instance of epsilon
- Deploy that instance beside the existing instances
- Adjust the HAProxy configuration to redirect traffic
- First staging environment than production



PLAN 4/5

Decommission OpenID

- Remove the running instance
- Adjust HAProxy



PLAN 5/5

PROFIT!!!

10

REALITY 1/8



11

REALITY 2/8

Migration of apps to OIDC

- More work than anticipated
- Apps outside Fedora Infrastructure are using Fedora OpenID
- Few apps were rewritten from scratch
- It took as 3+ years to get ready

REALITY 3/8

Announcement – we are finally working on it!

- 20th March 2025
- [Community blog post](#)
- We are working on getting rid of OpenID finally!

REALITY 4/8

May update – still struggling

- 13th May 2025
- [Community blog post](#)
- Still apps that are using OpenID
- Started working on ipsilon instance separation

REALITY 5/8

Separate the OpenID instance

- Lack of OpenID apps
 - Have to create reproducer (python 3.6!)
- Separating config for OpenID
- Some changes in HTTPD config as well
- New VM for OpenID instance
- Added banner to inform about date of sunset

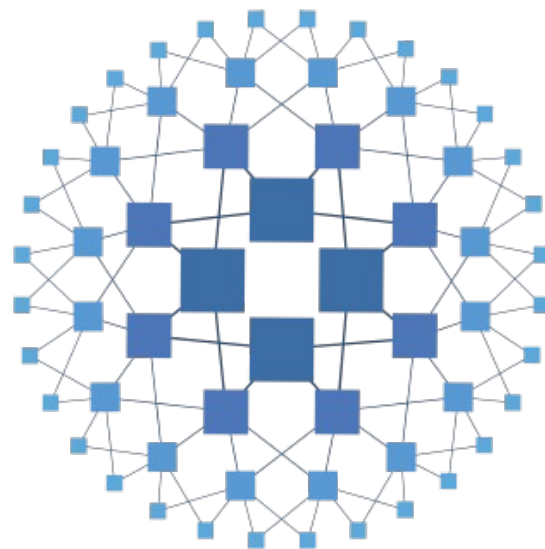


15

REALITY 6/8

HAProxy changes

- Difficult
- XRDS (**e**xtensible **r**esource **d**escriptor **s**equences) endpoint needs to be provided by other instance
- Persistent cookie to know where to redirect the request
- Problem with certs for `<username>.id.fedoraproject.org`



HAPROXY

16

REALITY 7/8

November update – it's finally ready!

- 25th November 2025
- [Community blog post](#)
- OpenID instance separated
- Giving projects time to migrate if they didn't till now
- Two months till the separation of OpenID in ipsilon

REALITY 8/8

End of life date

- Originally **May 20th 2025**
 - Very optimistic
 - Some Fedora services still using OpenID
- Moved to **May 1st 2026**
 - Holiday!
- Real date **May 5th 2026**



CURRENT SITUATION

- The OpenID is gone
- We didn't see anybody screaming (it could still happen!)
- Users didn't notice the change at all
- Work being done to find replacement for epsilon

FUTURE

- Replacement of ipsilon with something better (keycloak, ipa-tuura)
- More stable authentication
- Getting rid of tech debt



20

BONUS: INTRODUCING NEW OIDC SERVICE

- Open a ticket on Fedora infrastructure
- Provide us with required information
 - URL of the application
 - Redirect URL for OIDC authentication
 - Name of service
 - Contact
- We will reach to you with `client_id` and `client_secret` to use
- Best to test on staging first if possible

THANKS!

